

ZEYU CHEN

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>> SKILLS

Languages: Python, C++, JavaScript, Java, SQL, HTML, CSS

Frameworks & Libraries: OpenCV, Numpy, Django, React, Jinja, Apollo GraphQL, Tailwind, Leap Motion SDK

Tools & Technologies: Git, Vim, Google Cloud Platform (GCP), Unix / Linux

>> PROJECTS

Colonel By Online Judge Programming Platform (cboj.ca)

July 2020 - Present

- Extended, built and open-sourced a custom web platform deployed for a high school computer science curriculum for assessments and evaluations using **Python**, **Django**, **Jinja**, **HTML** and **CSS** — reaching **90+ daily users** in 2021
- Evaluated various cloud IaaS; hosted with **Google Cloud Platform (GCP)** Compute Engines expensed by the school
- Performed monitoring with **Bash** scripts on **Unix** environment to meet teachers' performance specifications
- Redesigned additional privacy levels with custom point-tallying using **Python** and **MySQL** for student assessments
- Implemented a user-friendly portal for teachers to upload customized verification scripts

3D Chess

May - June 2021

- Built a three-dimensional variant of chess using **JavaScript**, **WebAssembly** and **C++** with a team of 3
- Designed an automated player with varying difficulties by implementing **Minimax** search with alpha-beta pruning
- Reduced the search runtime by **over 70 seconds** per turn using pruning methods, heuristics and optimizations

Mole-croscope (CuHacking 2nd Place, Innovapost Challenge Winner)

Feb 2019

- Collaborated with team of 4 to build a mole-tracking **Android** app with skin cancer prediction trained on **GCP**
- Designed mole difference-tracking feature by implementing object translation invariance with **Python** and **OpenCV**
- Implemented **Python** preprocessing scripts to auto-correct skew and translation differences between user inputs

>> EXPERIENCE

Contest Organizer and Moderator :: DMOJ: Modern Online Judge

July 2020 - Present

- Moderated Canada's largest online judge, dmoj.ca, with over 100 000 users, 300 contests, and 3 500 problems
- Co-authored and organized 7 programming contests averaging ~200 participants each, as part of a team of 5-6
- Designed **graph theory**, **data structures** and **dynamic programming** problems and published solution editorials
- Ensured fair point allocation across tier levels by implementing solutions approaches and designing test cases

>> AWARDS

- Canadian Computing Olympiad (CCO 2021) :: **Silver Medalist**
- Canadian Computing Competition (CCC 2021) :: **Perfect Score, top 1.1%** (~3 000 participants)
- Facebook Hacker Cup :: **Top 1.2%** in 2020, **top 1.6%** in 2021 (~35 000 participants)
- Google Kickstart Competition 2021 :: Round C **top 0.4%**, 48th worldwide (~12 200 participants)
- Google Kickstart Competition 2020 :: Round G **top 0.9%**, 73rd worldwide (~8 100 participants)
- Advent of Code :: **Top 0.2%** in 2018, **top 0.2%** in 2019, **top 0.03%** in 2020 (~100 000 participants)
- USA Computing Olympiad 2019, 2020 :: **Platinum division**

>> EDUCATION

University of Waterloo :: B. of Software Engineering

Sept 2021 - Present

Candidate for Bachelor of Software Engineering, Honours, Co-operative Program

Expected graduation April 2026

GPA: 3.93 / 4